2024 3rd International Conference for Advancement in Technology (ICONAT)
Goa, India. Sep 13-14, 2024

A Conceptual Framework for Smart Agriculture Management using Convolutional Autoencoder with Long Short-Term Memory

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***Abstract*— Recently, smart agricultural farming is emerging with high improvements in practical applications, making it a very significant valuable, and important device. The identification of plant disorder is one of the significant aspects of managing an agriculturally implemented nation. The effective and timely identification of plant disorders is significant for a productive and healthy agricultural domain and eliminates wasting money and other significant resources. The protection of crops plays an important role in enhancing requirements for food quantity and quality. The traditional crop yield prediction demands plant acreages along with sample validations that require plant-cutting research. Because of the poor learning ability of conventional machine learning-aided irrigation techniques, these models are not applicable to unpredictable climates. Hence, a new smart agriculture management model is presented using deep learning to perform crop yield detection, irrigation, crop detection, and disease detection. Initially, the data and images needed for smart agriculture management are gathered from benchmark sites. At first, the collected data is utilized to perform crop yield detection and irrigation via the developed Convolutional Autoencoder with Long Short-Term Memory (CAE-LSTM). This process is helpful to decrease the water usage in crop fields and increase the crop production. Then, the accumulated images are utilized by the proposed CAE-LSTM for performing crop detection and crop disease detection. The proposed deep learning-assisted smart agriculture management approach is useful in diagnosing crop disease at its early stages. At last, the superiority of the suggested system is evaluated with existing techniques to analyze its performance.**

Keywords*— *Smart Agriculture Management; Crop Type and Disease Classification; Smart Irrigation; Crop Yield Prediction; Convolutional Autoencoder with Long Term Short Memory

I. Introduction

Smart agriculture has the ability to provide numerous outcomes to the revolution of agriculture. In modern times, agriculture is encountering issues that may affect its future including productivity, crop quality, drought, and yield prediction issues [1]. In the past years, several techniques have been developed to rectify the issues and impediments that occur with smart farming including crop productivity, species identification, yield prediction, disease identification issues, drought, and the management of irrigation [2]. There are some experiments in employing data evaluating techniques for effective decision support models for forming information. The majority of the experiments face customary mechanisms without noticing the functionality of the technique, and several of them ignore the data preprocessing in the starting phases [3]. The incorporation of IoT in agriculture is the primary solution because there is no demand for periodical control and monitoring in this sector. With the implementation of IoT, the applications of smart agriculture have been implemented highly in the modern days [4]. In IoT, the Wireless Sensor Network (WSN) gathers the data from the sensors installed at numerous nodes and transfers the data over the wireless protocol, a vast amount of data from the IoT agriculture model is gathered including humidity, wind speed, and temperature that offers the data regarding the environmental attributes, allowing the climate predictions [5]. Precision agriculture has high power and can make important contributions to security, food production, and safety.

The accurate long-term and short-term forecasting of future weather situations can support the policymakers, distributors, and farmers to make decisions for better production in agriculture that improves the activities, which enhances the advantages of usage and implementation of food resources for economic, social, and individual purposes [6]. The evaluation and design techniques of future weather can support the development of possible management in agriculture, hence supporting and assisting the production management towards better agriculture development [7]. Forecasting on the basis of weather information is a complex issue due to the gathered data from the sensors including difficult non-linear relationships with numerous factors. Whereas, the high sampling frequency and the huge data that have been gathered and saved in the IoT device make it applicable to evaluate the sensory information, find out new data, and forecast future perceptions.

Several techniques have been recommended to rectify the prediction issues for the gathered time-sequential data on the basis of the IoT system's sensor [8]. For example, the conventional Echo State Network (ESN), Support Vector Machine (SVM), and Artificial Neural Network (ANN) with swarm algorithms have been utilized in the designing and forecasting of the time sequential data's future [9]. But, practically for the IoT device, these techniques cannot attain

2024 3rd International Conference for Advancement in Technology (ICONAT) | 979-8-3503-5417-1/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICONAT61936.2024.10774647

correct predictions because of the difficulty in the gathered data and the weak designing capacity for nonlinearity. Nowadays, deep learning techniques have displayed high advancements for difficult nonlinear data feature extraction [10]. Especially, the "Recurrent Neural Network (RNN), and the Convolutional Neural Network (CNN) and its enhanced techniques including LSTM, Bi-directional LSTM (Bi-LSTM), and Gated Recurrent Unit (GRU)" have been employed for obtaining the time sequential information features. But, the functionality of these models still requires to be implemented to attain very correct prediction for the weather information to satisfy the demand for proper agriculture.

The implemented smart agriculture management system contains the below contributions.

- ✓ To implement a new agriculture management system by utilizing IoT and deep learning that helps to classify the crop types and crop diseases for making a high profit. Moreover, it supports to predict crop yield and irrigation for minimizing financial losses.
- ✓ To design a CAE-LSTM technique by integrating the CAE with the LSTM that supports to improve agriculture management by making accurate predictions and classifications. This relatively supports the farmers to make timely and correct decisions in the agricultural field.
- ✓ To investigate the performance of the implemented smart agriculture management system by utilizing numerous existing classifiers.

The developed smart agriculture management model contains the following divisions. Division II includes the conventional agriculture management models. Division III displays an effective framework of agriculture management using deep learning. Division IV provides the structural and conceptual representation of CAE with LSTM in the agriculture field. Division V gives the results and discussions of the implemented system. Division VI provides the conclusion of the implemented system.

II. Existing Works

A. Related Works

In 2022, Shaikh *et al.* [11] have explained the significance of modern technologies in conventional agriculture and the complexities that developed when these techniques were employed in farming mechanisms. The authors performed a detailed experiment and explained the current and future innovations in Artificial Intelligence (AI) and addressed the conventional research issues.

In 2020, Jin *et al.* [12] have recommended a hybrid deep learning model that was employed to decompose the climate information with distinct frequency features, further a GRU technique was trained. Further, the outcomes from the GRU were added to attain the prediction outcome. The analysis on the basis of climate data from IoT devices validated the implementation of the recommended model. The solutions displayed that the model could attain very accurate predictions.

In 2020, Jin *et al.* [13] have employed the deep learning predictor in that the weather data was decomposed and further, the GRU was trained. In the end, the outcomes from GRU were integrated to achieve the long-term and short-term prediction outcomes. The research was validated for the recommended model in that the prediction solutions displayed that the designed process could achieve accurate prediction and satisfy the requirements of accurate production in agriculture.

In 2021, Kashyap *et al.* [14] have implemented a deep learning-aided IoT-based smart irrigation process for the precision agriculture. The model employed LSTM to forecast the volumetric soil moisture information for one day before, irrigation time, and so on. It was proof from the simulation outcomes that the designed model employed water more wisely than the conventional models.

In 2021, Rezk *et al.* [15] have developed an intelligent technique utilizing the wrapper feature selection mechanism and further provided a classification mechanism that was recommended for drought and crop productivity prediction. The outcomes displayed that the implemented model was precise, accurate, and robust to categorize and forecast the drought and cop productivity. From the outcomes, the recommended model was confirmed to be very accurate contrasted to the existing mechanisms.

B. Research Gaps and Challenges

Nowadays, smart agriculture has achieved high focus because of the enhanced population requirements for water and food. The farmers demand arable land and water sources for rectifying these demands. Because of the lack of significant resources, the farmers demand a solution that modifications the way they perform. A significant experimental direction of the IoT model for accurate agriculture is to offer a particular environment for profit. Some of the recent innovations such as machine learning and deep learning models are also integrated with the IoT for performing precision agriculture. However, the existing agriculture management models couldn't provide satisfactory solutions. Table 1 gives some conventional agriculture management technique's advantages and limitations. Additionally, several significant research gaps in the conventional models have been provided below.

- Smart agriculture management is important for producing high crop yields and also minimizing financial loss. Though there are numerous strategies have existed in the past years, these strategies face some notable problems such as inaccuracy because the sensing data are nonlinear, complex, and include numerous components. Hence, an effective smart agriculture management model is necessary.
- Some smart agriculture management models only focus the crop disease classification and crop yield prediction. However, these techniques didn't consider the crop type classification and smart irrigation processes. In order to make accurate decisions, performing all these kinds of processes is necessary that make a better agriculture management model.
- Processing both plant images and the data obtained from the IoT sensors helps to make

better agriculture management; however, the traditional techniques fail to consider that. Hence, implementing a new agriculture management system is necessary.

- Though some existing techniques utilize deep learning models, these techniques don't extract the necessary features while processing more input images and data. Hence, an effective strategy is required for extracting the significant features from the given data and images for making better agricultural decisions.

Therefore, an innovative smart agriculture management system is presented for rectifying the limitations in the existing systems.

TABLE I. FEATURES AND CHALLENGES OF EXISTING SMART AGRICULTURE MANAGEMENT MODELS

Author [citation]	Methodology	Features	Challenges
Shaikh *et al.* [11]	Machine learning	• It is very effective and performs the automated tasks. • It produces accurate solutions.	• It has very poor interpretability • Its computational cost is high.
Jin *et al.* [12]	CNN	• It demands no manual supervision. • It is capable to handle huge data sources.	• Its computational requirements are high. • It utilizes more training time.
Jin *et al.* [13]	GRU	• It is a very powerful technique that captures long-range frequencies. • It provides highly accurate solutions.	• It couldn't perform well on small data sources. • It is susceptible to overfitting.
Kashyap *et al.* [14]	LSTM	• It can manage a huge amount of input features. • It can manage the long term sequential information.	• It demands more training data. • It is a very complicated network.
Rezk *et al.* [15]	Wrapper method	• It minimizes the computational complexity. • It increases the interpretability and accuracy.	• It is prone to overfitting. • It takes more time for the training process.

III. AN EFFECTIVE FRAMEWORK OF AGRICULTURE MANAGEMENT USING DEEP LEARNING

A. Input Sources

The implemented smart agriculture management system employs the following data sources.

Dataset 1 ("PlantifyDr Dataset"): Using the link "https://www.kaggle.com/datasets/lavaman151/plantifydr-dataset: access date: 2024-07-01", this resource is attained from the Kaggle platform. This resource includes 1, 25,000 jpg images and also this resource includes ten distinct plant categories. Thirty-seven plant diseases are taken into account in this dataset.

Dataset 2 ("Crop Yield Prediction Dataset"): From the Kaggle platform, using "https://www.kaggle.com/datasets/patelris/crop-yield-prediction-dataset?select=yield.csv: access date: 2024-01-07", this resource is fetched. It is employed to forecast the crop yields of the most utilized crops in the world. The files are in the format of .csv and include the data regarding the weather conditions, the history of crop yield, and the pesticides.

From the aggregated data sources, the obtained images are pointed as p_x, $here\ x = 1,2,...,X$ and the gathered data is given as q_y, $here\ x = 1,2,...,Y$. Here, the attributes $X\ and\ Y$ denote the overall count of the image and data accordingly. The sample images of the designed smart agriculture management process are given in Fig.1.

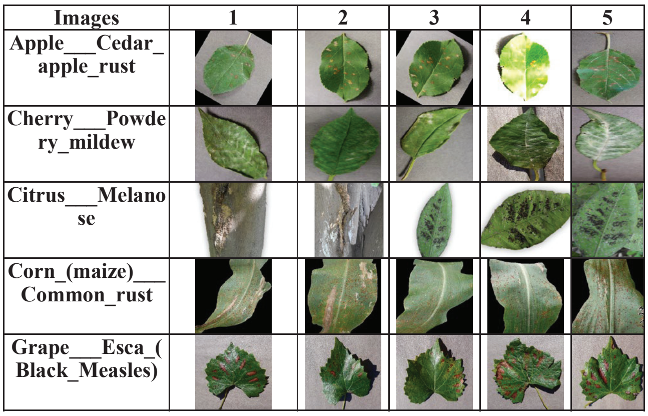

Fig. 1. Sample images of the designed smart agriculture management system

B. Proposed Model and its Description

Smart agriculture supports the farmers to make more profit and make sure the country's economy is enhanced. Smart agriculture utilizes a mechanism named precision farming, where the overall aspects of the environment needed by the crop to grow are periodically observed. Observing alone cannot support the crop's health and hence maintaining these aspects if possible also needed. Additionally, these data are stored and utilized for the future prediction of the appropriate plant to be grown in a specific environment. The traditional technique includes huge manual intervention, which is prone to error and demands more time. Hence, there is a demand for smarter ways of crop production utilizing machine learning and IoT. The ideas of "IoT, machine learning, and cloud computing" are utilized to implement the solution to this problem. However, conventional machine learning techniques can't process a huge amount of data in real-time accurately. Therefore, a deep learning-aided smart agriculture management system is required.

The primary goal of the implemented model is to design smart agriculture management with the help of deep learning, where crop detection, disease prediction, yield prediction, and irrigation tasks are performed. At first, the necessary images and data are gathered from the standard datasets for smart agriculture management. With the support of the gathered data, the crop yield detection and irrigation tasks are performed using CAE-LSTM. This operation is supportive to minimize water usage in the crop yields and improve the production of crops. Further, the aggregated images are employed to perform the crop type and disease prediction with the support of the same CAE-LSTM technique. With the support of a designed smart agriculture management model, crop diseases are diagnosed timely. Finally, the effectiveness of the implemented model is estimated with conventional techniques using various analyses.

IV. Structural and Conceptual Representation of Convolutional Autoencoder with LSTM in Agriculture Field

A. Developed CAE-LSTM

The CAE-LSTM technique is implemented for classifying the crop type and disease and predicting the irrigation and crop yield. The designed CAE-LSTM approach is a combination of the existing CAE and LSTM. The functionalities of these two techniques are given as follows.

The CAE [16] is the improved variation of the traditional AE. In this technique, the fully connected layer is modified with the layer of convolution. This model has the merit of AE's both convolutional layer and the pre-trained capacity. When compared with the existing autoencoder, the improved CAE includes a convolutional layer for both decoder and encoder sections. But, in the CAE, the fully connected layer is included in the decoder section. This model minimizes the cost of computation. Eq. (1) shows the expression of convolution, deconvolution layer with the activation function.

$$e^r = \sigma\left(\sum_{t \in T} u^t \otimes y^r + p^r\right) \tag{1}$$

Here, the current layer's latent indication is taken as e^r, and the activation function is indicated as σ. The current layer's weight is given as u^t and the 2D convolution layer is taken as $\otimes$. The conventional layer's feature map is given as y^r. The current layer's bias is given as p^r.

The LSTM [17] is motivated by the Recurrent Neural Network (RNN) and has the capacity to learn the long dependencies among the factors. Normally, the LSTM has three functional gates such as output, input, and forget. By changing the gates of LSTM, this model employs temporary memory. This model improves the gradient disappearance and the explosion. The overall computation is derived from Eq. (2) to Eq. (5).

$$p_{(x)} = \sigma\left(L_{pt} t_{(x)} + L_{pf} f_{(x-1)} + v_p\right) \tag{2}$$

$$q_{(x)} = \sigma\left(L_{qt} t_{(x)} + L_{qf} f_{(x-1)} + v_q\right) \tag{3}$$

$$r_{(x)} = \sigma\left(L_{rt} t_{(x)} + L_{rf} f_{(x-1)} + v_r\right) \tag{4}$$

$$w_{(x)} = \tanh\left(L_{wt} t_{(x)} + L_{wf} f_{(x-1)} + v_w\right) \tag{5}$$

Here, the bias vector is indicated as v. The weights are specified as L_t *and* L_f. The activation function is given as σ. Then the non-linear activation function is $\tanh$.

The LSTM technique's hidden state and the memory cell are upgraded using Eq. (6) and Eq. (7).

$$w_{(x)} = p_{(x)} \bullet w_{(x)} + q_{(x)} \bullet w_{(x)} \tag{6}$$

$$f_{(x)} = r_{(x)} \bullet \tanh\left(w_{(x)}\right) \tag{7}$$

Here, the point-wise multiplication operation is given as $\bullet$.

CAE-LSTM: The CAE technique helps to reconstruct the image's missing image and also minimizes the high dimensionality of the image. Moreover, this technique effectively selects the rich features from the both data and the image. The LSTM technique utilizes historical data for making better predictions and also provides solutions quickly. However, the LSTM technique can't extract the necessary features when executing vast amounts of data or images. This directly affects the accuracy of the overall process. Hence, the CAE technique is integrated with the LSTM model for providing more accurate solutions. The CAE effectively extracts the rich and necessary features from both data and image thus improving the accuracy of the process. The CAE-LSTM model's structure is shown in Fig.2 for providing smart agriculture management.

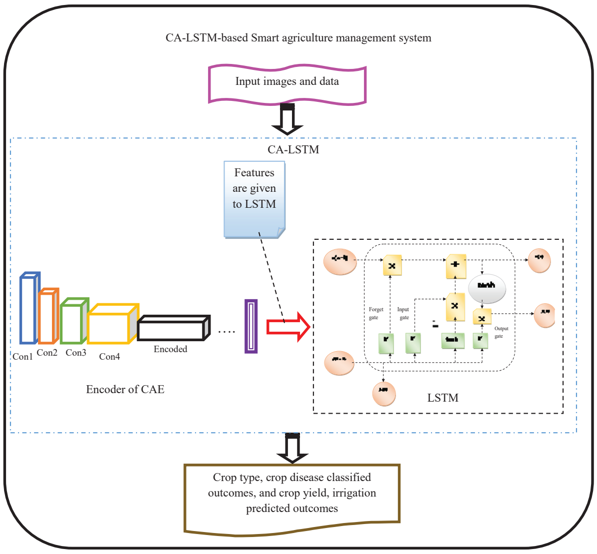

Fig. 2. Implemented CAE-LSTM technique's structure for smart agriculture management system

B. Four Different Objectives for Effective Management using CAE-LSTM

The implemented smart agriculture management system includes four primary objectives including crop type classification, crop disease classification, irrigation prediction, and crop yield prediction. These four objectives are performed by the developed CAE-LSTM technique. These objectives are explained as follows.

Crop type classification: The classification of crop type offers important data for numerous decision-making tasks needed to manage the resources in agriculture. Reliable data should be available on crops hence that agricultural management can be enhanced and expenses can be minimized. Hence, in this smart agriculture management, the crop type classification is performed. In this process, the gathered images p_x are given to the developed CAE-LSTM model. Here, the CAE technique extracts the necessary features from the given images and the resultant features are given to the LSTM technique. Finally, the crop type classified outcome is achieved for smart agriculture management using CAE-LSTM.

Crop disease classification: The crops are sensitive to distinct diseases during cultivation. The diseases affect the growth of the crops. The timely and accurate disease classification can minimize the possibility of additional damage to the crops. The issues can develop when trying to forecast the crop disease types manually. Hence, the classification of crop disorder is performed automatically in this developed model. Here also, the gathered images p_x are fed into the CAE-LSTM method for classifying the crop disease. Here, the CAE is responsible for extracting the important features from the given images, and the features are forwarded to the LSTM model. Finally, the crop disease classified outcomes are achieved.

Irrigation prediction: High consumption of water in agriculture affects the safe access of numerous people to drinking water. Smart irrigation supports very effective utilization in irrigated agriculture. To determine the irrigation demands of numerous soil texture classes in the production field demands much more effective strategies. Therefore, a smart irrigation prediction process is performed in this implemented model. In this process, the collected field data q_y are forwarded to the CAE-LSTM approach for making better predictions. The CAE extracts the requisite features and the features are fed into the LSTM model for making predictions. Thus, the irrigation-predicted outcomes are attained.

Crop yield prediction: The crop yield prediction becoming very important owing to the enhancing concern regarding the security of food. The earlier crop yield prediction plays a significant role in decreasing food scarcity by validating the food availability for the growing population. Hence, the crop yield prediction is performed in the designed agriculture management system. In this operation, the aggregated field data q_y is taken as input for the implemented CAE-LSTM approach. The important features are obtained with the support of CAE and the features are given to LSTM for making accurate predictions. Finally, the crop yield predicted outcomes are achieved.

Thus, an effective smart agriculture management system is implemented in this work by the CAE-LSTM model.

V. Results and Discussions

A. Experimental setup

The implemented smart agriculture management system was developed with the support of the Python platform and promising results were obtained. By analyzing the classification process, some traditional models including "VGG16 [18], ShuffleNet [19], CAE [16], and LSTM [17]" were utilized. Besides, for analyzing the prediction process, several conventional techniques including "Deep Neural Network (DNN) [20], 1D-Convolutional Neural Network (1D-CNN) 1D-CNN [21], CAE [16], and LSTM [17]" were utilized.

B. Performance evaluation of implemented crop type classification process

The designed crop type classification operation's performance is analyzed in Fig.3 with the support of some specific evaluation factors over traditional techniques. By varying the values of the learning percentage, this experiment is conducted. For the 45th learning percentage, the designed crop type classification model's accuracy is improved by 10.52% of VGG 16, 8.42% of ShuffleNet, 5.26% of Autoencoder, and 3.15% of LSTM accordingly in Fig.3 (a). Thus, the implemented crop type classification task achieved improved accuracy than the existing models.

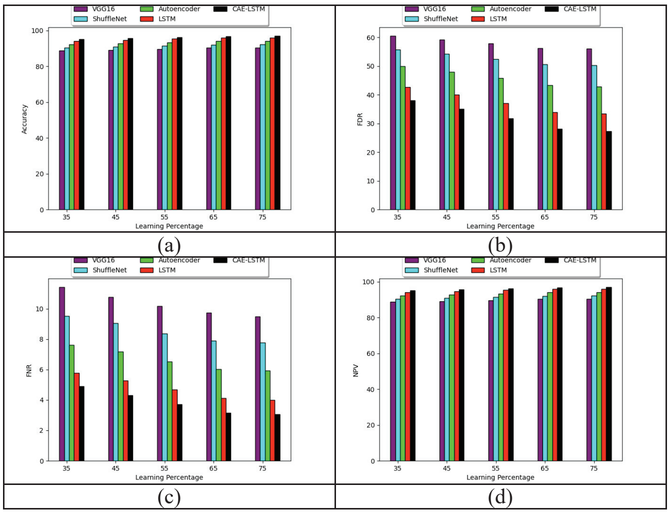

Fig. 3. Performance investigation of developed crop type classification model over traditional models in terms of " (a) Accuracy, (b) FDR, (c) FNR, and (d) NPV"

C. Performance evaluation of implemented crop disease classification process

The designed crop disease classification process's performance is investigated in Fig.4 over some conventional methods. The epoch counts are supported for analyzing the performance in this experiment. For the 200th epoch, the developed crop disease classification framework's FPR is decreased by 36.2% of VGG 16, 52.9% of ShuffleNet, 21.4% of Autoencoder, and 17.7% of LSTM respectively in Fig. 4 (c). Thus, the developed crop disease classification process achieved very low error rates than the conventional techniques.

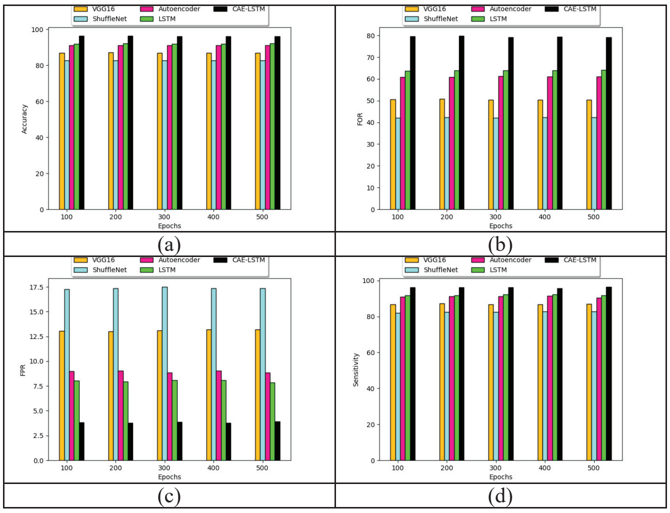

Fig. 4. Performance investigation of implemented crop disease classification model over traditional models in terms of " (a) Accuracy, (b) FOR, (c) FPR, and (d) Sensitivity"

D. Performance evaluation of implemented irrigation prediction process

The designed irrigation prediction approach's performance is validated in Fig.5 over existing classifiers. The epoch values are varied for validating the irrigation process's performance. For the 100th epoch value, the MASE of the designed irrigation prediction task is decreased by 76.4% of DNN, 61.7% of 1DCNN, 41.1% of Autoencoder, and 38.2% of LSTM appropriately in Fig.5 (a). Thus, the implemented irrigation process accomplished more effective solutions than the conventional methods.

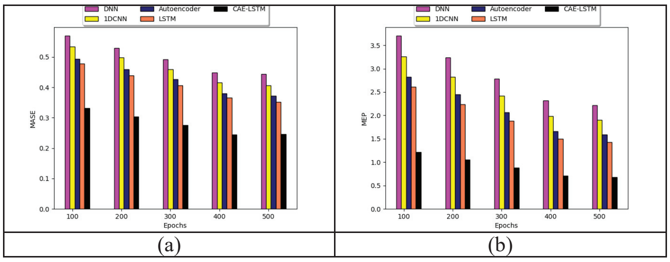

Fig. 5. Performance analysis of implemented irrigation prediction process over conventional classifiers concerning " (a) MASE, and (b) MEP"

E. Performance evaluation of implemented crop yield prediction process

The implemented mechanism's performance is examined in Fig.6 over traditional models. By utilizing the activation functions this experiment is performed. For the ReLU activation function in Fig.6 (b), the RMSE of the designed task is minimized by 74.6% of DNN, 55.5% of 1DCNN, 34.9% of Autoencoder, and 11.1% of LSTM correspondingly. Thus, the developed crop yield prediction process obtained higher efficiency than the other classical techniques.

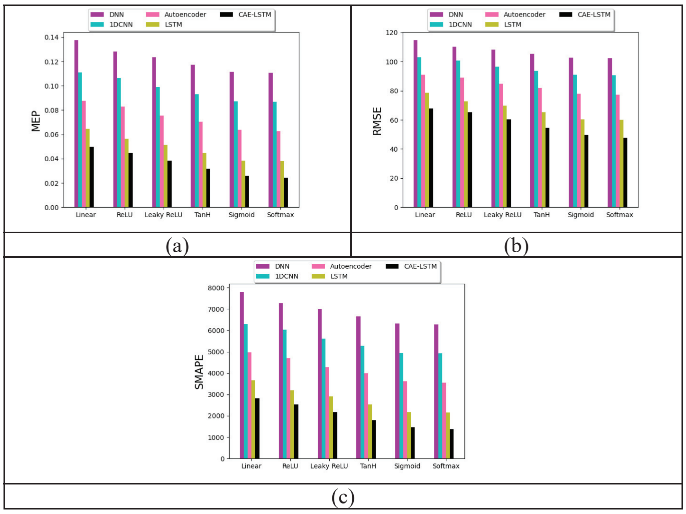

Fig. 6. Performance evaluation of implemented crop yield prediction process over existing models concerning " (a) MEP, (b) RMSE, and (c) SMAPE"

VI. CONCLUSION

The developed work has presented a novel agriculture management system, where the here crop detection, disease prediction, yield prediction, and irrigation operations were conducted utilizing deep learning. From the benchmark sites, the required data and the images were garnered for the implemented agriculture management model. The gathered data was given to the designed CAE-LSTM technique for performing crop yield prediction and irrigation. This task was helpful to reduce water usage in the crop yields and enhance the production of the crop. After that, the gathered images were given to the same CAE-LSTM model for performing the crop type and disease prediction. The crop diseases were diagnosed at early times with the support of the developed smart agriculture management model. Lastly, using various evaluation measures the performance analysis was conducted. The implemented crop disease classification process's accuracy was increased by 10.52% of VGG 16, 15.78% of ShuffleNet, 5.26% of Autoencoder, and 4.21% of LSTM respectively for the 400th epoch. From this experiment, it has been confirmed that the implemented smart agriculture management produced highly accurate solutions for better decision making processes. The implemented smart agriculture management model provided more effective solutions. In the future, a recent optimization algorithm will be integrated with the deep learning model for minimizing the computational time in the developed smart agriculture management.

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